

AANSH MALHOTRA

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SUMMARY

GenAI-focused Full-Stack Developer building AI-powered web applications using LLMs, RAG, modern frontend and backend technologies. Strong in turning GenAI capabilities into practical user-facing products while handling the underlying APIs, data, and application workflows. Seeking a GenAI or Software Development internship to build and ship real-world AI-powered products.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript

GenAI: OpenAI SDK, Google Gemini SDK, Anthropic SDK, RAG, AI Agents, AI Workflows, Prompt Engineering

Frontend: Next.js, React.js, Tailwind CSS, Shadcn UI, Zustand, TanStack React Query, Context API

Backend: Node.js, Express.js, REST APIs, Auth, Socket.IO, SSE

Databases & Caching: PostgreSQL (Neon), Prisma, Qdrant, MongoDB, Mongoose, Redis

Cloud & DevOps: Docker, Docker Compose, Vercel, Render

Tools & Integrations: BullMQ, Inngest, Git, GitHub

PROJECTS

ChatSource – AI Knowledge Assistant | [GitHub](#) | [Live](#)

- Built an **AI knowledge assistant** that lets users add PDFs, text, websites, and YouTube videos and ask questions across their sources instead of manually searching through lengthy content, with answers supported by relevant citations.
- Implemented a **RAG pipeline** with embeddings, multi-query retrieval, result fusion, and reranking, along with source filtering and asynchronous processing using BullMQ and Redis for parsing, chunking, embedding, and indexing content.
- **Tech Stack:** Next.js, TypeScript, Node.js, Express.js, PostgreSQL, Prisma, Qdrant, Redis, BullMQ, OpenAI, Google Gemini, Clerk

Raseo SDK – TypeScript AI Agent SDK | [GitHub](#) | [Live](#)

- Built a **TypeScript SDK for building AI agents and LLM applications** with a unified interface across OpenAI, Anthropic, and Google Gemini, allowing developers to switch model providers without changing application-level agent logic.
- Implemented **provider abstraction, real-time streaming, type-safe tool calling, Zod-based schema validation, and an automated agent reasoning loop** for multi-turn interactions, tool execution, and structured AI workflows.
- **Tech Stack:** TypeScript, Node.js, OpenAI, Anthropic, Google Gemini, Zod

Chatify – Real-Time Scalable Chat Application | [GitHub](#) | [Live](#)

- Built a **real-time messaging application** supporting instant messaging, online/offline presence tracking, and persistent chat history using a scalable REST and WebSocket-based backend architecture.
- Implemented **backend performance optimizations** including **cursor-based pagination, MongoDB indexing, and Redis-based multi-level rate limiting** with 30 messages/minute per user and 15 messages/10 seconds per user per conversation to prevent spam and API abuse.
- **Tech Stack:** MongoDB, Express.js, React.js, Node.js, Socket.IO, Redis, Zustand, JWT, Tailwind CSS

OPEN-SOURCE EXPERIENCE

GirlScript Summer of Code (Tech Contributor)

- Merged **39 pull requests** across multiple open-source repositories, contributing new features, bug fixes, and UI improvements through GitHub-based development workflows.
- Built features including **Certificate Management, Project Gallery, and GitHub API integrations** using React.js, JavaScript, and Tailwind CSS.
- Collaborated with project maintainers through issues, pull requests, code reviews, and iterative changes across community-driven projects.

EDUCATION

Guru Nanak Dev University, Amritsar

B.Tech - Computer Science and Engineering - CGPA - 7.88

2024 – 2028

Amritsar, Punjab